

Staff (teacher) portal: workflow audit

Huchu schools vertical audit · Staff (teacher) portal: workflow audit · staff-portal/workflows.md · rendered 2026-09-15

Surface: `staff.<tenant>` host, routed to `app/portal/teacher/**`, backed by `app/api/v2/schools/portal/teacher/**`, `lib/schools/teacher-day-loader.ts`, `teacher-day.ts`, `teacher-reports.ts`, and the shared routes under `/api/v2/schools/{assessments,assignments,meetings,notices,syllabus,teaching-resources}`. Audience: staff with a `SchoolTeacherProfile` whose `userId` is the signed-in User of role `TEACHER` (HOD and WARDEN have no portal home today).

Audit date: 2026-09-14. Method: static read of the code on `main`, cross-checked against the roadmap, the prototype `docs/design-system/portals/teacher.html`, and the K-12 benchmark. Companion UI/UX document: `ui-ux.md` in this folder.

Note on naming: the docs call this the teacher portal; the user's brief calls it the staff portal. The host prefix is `staff.` (`lib/platform/portal-hosts.ts:29-34`), which is the better name, because non-teaching staff, HODs and wardens will need a home too. This document uses "staff portal" for the surface and "teacher" for the persona.

1. Verdict in one paragraph

The staff portal is the most complete of the three portals and the roadmap's own priority. Today, register capture, assessment marks, marks book with roll-up, homework authoring and marking, lesson planning with copy-forward, timetable with cover shown, link-only resources, reports, meetings, and message replies all work and are scoped to the teacher's own assignments. It fails its user in four places that matter daily. A register saved here never leaves DRAFT because there is no submit, so the office board and the parent app both show every class as unsubmitted. Two prototype actions (open a parents' evening, "tell the family" from reports) always 403 because the TEACHER persona lacks the grant the shared route checks. The result-sheet chain after roll-up (submit, HOD moderation, publish, edits after rejection) lives only in the admin workspace, so a teacher cannot see where their marks are or why they came back. And the term-marks roll-up route is permission-checked but not ownership-checked, so a teacher can roll up any class. Eight stories are open; three of them (messaging, custom columns, publish to parents) are the ones teachers ask for first. Since this was written the register can be sent in from the portal, the two 403s are gone at the persona rather than at the button, the roll-up and the class reads are scoped to the teacher's own assignments, the office queue is no longer every teacher's to read, and a teacher can create an assessment and start a conversation with a family. The result-sheet chain after roll-up is still absent from the portal, the resource library still takes links only, there is still no offline register, and the HOD still has no portal home.

Runtime check (14 September 2026)

Verified on the seeded St Marys tenant as `wellington.mabika@stmarys.test` (role TEACHER, Business Studies and Physics, see `../reference/runtime-verification.md`). Confirmed at runtime: B1 (saving a 20-pupil register returns a session with status DRAFT and the screen has no submit control), B2 (`POST /meetings` returns 403 "Your role cannot edit students"), B3 (`POST /notices` returns 403 "Your role cannot create reports"), B4 (term marks and assessments for a class this teacher does not teach return 200), and the

absence of any start-thread or broadcast control on Messages. Inconclusive: the bell badge (this teacher had no papers to mark).

Revised on 15 September 2026, after the eighteen commits that implemented against this audit. A finding that is now closed keeps its description and gains a sentence saying what closed it; a finding this audit got wrong says so and gives the true position; everything else stands as written.

2. Docs versus code

Claim	Source	Code reality	Verdict
S-6.41 register with mark-all and per-pupil override: <code>done</code>	roadmap	Save works (<code>me/attendance</code>). No submit; session stays DRAFT (<code>route.ts:106-118</code>). Only the office can submit or lock. Fixed: the screen writes the marks, takes the session id off that response and calls <code>POST /attendance/sessions/[id]/submit</code> , so the day leaves DRAFT from the portal. Locking stays the office's, deliberately.	Was half the workflow. The acceptance signal never mentioned submit, which is how it passed.
S-6.43 gradebook cell by cell with autosave: <code>done</code>	roadmap	<code>teacher-marks-screen.ts</code> x saves scores per assessment; cannot create an assessment from the portal. Fixed: mark entry creates one, which the API always allowed a teacher to do.	True.
S-6.45 papers-to-mark queue: <code>done</code>	roadmap	Present on Today and Homework.	True.
S-6.56 parent meetings: <code>done</code>	roadmap	Read works; open evening and release slot POST to <code>/api/v2/schools/meetings</code> which requires <code>schools.students.edit</code> . TEACHER holds <code>view</code> only. Fixed: the route takes a new <code>book-meeting</code> verb, narrower than <code>edit</code> , and the teacher holds it.	Was marked done with the mutations broken; done now.
S-6.57 teacher reports: <code>done</code>	roadmap	Report loads; "tell the family" posts to <code>/notices</code> which requires <code>schools.reports.create</code> . TEACHER has no reports grant. Fixed: the family-notice path takes a new <code>notify-families</code> verb, which the teacher, the bursar and the head of department hold; writing a school-wide notice still needs <code>create</code> .	Was marked done with the action broken; done now.

Claim	Source	Code reality	Verdict
S-6.58 five settings panels: <code>done</code>	roadmap	Two preferences work; everything else is <code>NotYetAvailable</code> (<code>teacher-setting-s-screen.tsx:85-108, 384-410</code>). Half fixed: the eleven inert rows are gone and the prose says what the product cannot do yet. Two preferences still stand in for five panels.	Honest UI, wrong roadmap status; the UI is honest now without stamping eleven rows.
S-6.47 parent message threads: <code>todo</code>	roadmap	<code>/portal/teacher/message</code> s exists with reply; no start-thread UI. Settings copy still says messaging is not built (<code>:283</code>). Fixed: a teacher can start a thread with a family and write to a whole class, and the stale sentence is gone. Quick replies and attachments are still absent.	More built than the row says; the copy is fixed.
S-6.55 resource library with upload: <code>done</code>	roadmap	Links only; upload "arrives with the documents work" (<code>teacher-files-screen.tsx:404-408</code>).	Marked done, upload absent.
S-6.53 cover lessons: <code>todo</code>	roadmap	Cover is shown on the timetable; arranging cover is office-only by design (<code>me/lessons/route.ts:318-321</code>).	Accurate.
S-6.42 attendance history, S-6.44 custom columns, S-6.46 publish to parents, S-6.48/49 quick replies and broadcast, S-6.59 shared-device lock: <code>todo</code>	roadmap	Absent. Broadcast (S-6.49) is now built: a teacher can write to a whole class, one thread per guardian. Attendance history, custom columns, publishing to parents, quick replies and the shared-device lock are still absent.	Accurate then; S-6.49 is built.
Prototype: 14-item rail including Messages with unread count, Marks book at-risk, Reports cohort analytics, Security with 2FA and sign-out-all, Privacy with export and audit	<code>teacher.html</code>	Rail has the items; several panels are placeholders. The Messages unread count is real now and the placeholder rows are gone; Security and Privacy are still short of what the prototype draws.	Partial parity.
Help copy: "Access to pupil records is written to an audit trail"	<code>teacher-help-screen.tsx:222-228</code>	No portal route writes an audit row.	Copy contradicts behaviour.
<code>live-capabilities.md</code> : teacher portal is "attendance visibility, marks visibility"	system reference	Capture and entry both exist.	Stale doc.

3. Workflow inventory

#	Workflow	Status	Evidence and gap
T1	Get an account	Partial	No invite flow for teachers; office creates a <code>User</code> and a <code>SchoolTeacherProfile</code> with <code>userId</code> . Optional HR link via <code>teacher-identity.ts</code> .
T2	Sign in; shared-device sign-out; idle lock	Partial	Password sign-in works. Shared-device and auto-lock are <code>NotYetAvailable</code> . Signing out now asks first and says what an unsaved register costs; the inert shared-device and auto-lock rows are gone rather than built, and there is still no password reset.
T3	Today: lessons, right now / next, unmarked registers, marking pile	Implemented	<code>me/today</code> .
T4	Take the register	Implemented	Save to DRAFT only. No submit, no lock, no per-lesson register (keyed by class, stream, date), no offline queue. Fixed: saving writes the marks and the screen then submits, a register already sent in stays the teacher's to correct until the office locks it, and the three state questions live in one place instead of being written out inline in four routes. The per-lesson key (§9) and the offline queue (§8.6) are still open, and locking is the office's by design.
T5	Attendance history	Missing	S-6.42.
T6	Create an assessment	Implemented	<code>POST /assessments</code> allows TEACHER; no UI. Assessments are created in the admin class-results screen. Fixed: mark entry creates one, so a teacher no longer has to ask the office for the column they are about to fill.
T7	Enter marks for an assessment	Implemented	Absent flag supported.

#	Workflow	Status	Evidence and gap
T8	Marks book: term marks, roll up to sheet	Implemented	Roll-up not scoped to own class (§5). Fixed: ownership of the teaching assignment is checked, and because the roll-up clears the sheet and rewrites every subject's line it needs the whole class, so the control appears only for a moderator or for the one teacher who takes every subject in it. A teacher holding one subject of ten is told so before the press rather than by a 403 after it.
T9	Submit sheet, see HOD comments, fix after rejection, see publish state	Missing in portal	Routes exist under <code>/results/sheets/[id]/*</code> and are called only from admin screens. <code>me/marks</code> (edit after <code>HOD_REJECTED</code>) has no caller.
T10	Custom assessment column	Missing	S-6.44.
T11	Publish marks to parents under the school's rules	Missing	S-6.46; publishing is <code>SCHOOL_ADMIN</code> only and window-gated. The window deserves a second look: no seeded school configures one, and the report card refuses to print without an open window (B14). What a family reads in the portal is gated by the sheet's status and not by the window — a window check was written into the parent marks route during the implementation and deliberately reverted, because with no window configured anywhere it would have hidden every published mark from every family.
T12	Report-card comments, comment bank	Missing	<code>SchoolResultLine.remarks</code> writable via <code>me/marks</code> ; no UI.
T13	Set homework, publish, withdraw, mark	Implemented	Feedback text and send-back supported by API, not by the screen.
T14	Lesson plans, lay out week, copy last week	Implemented	
T15	Scheme of work	Read-only by design	Write needs <code>academics edit</code> (HOD or office).
T16	Resources: browse, add link, upload file	Partial	Links only.

#	Workflow	Status	Evidence and gap
T17	Timetable week with cover shown	Implemented	
T18	Request cover / report absence	Missing	Office arranges cover.
T19	Messages: reply to a family	Implemented	Office queue (<code>teacherProfileId: null</code>) visible to every teacher. Fixed: the office holds the unassigned queue, and a teacher sees only the entries about children they actually take. The rule that governs reading a thread now governs answering one.
T20	Messages: start a thread, quick replies, class broadcast, attachments	Partial	API <code>start</code> exists; no UI. S-6.48/49 and S-7.1. Fixed in part: a teacher can start a conversation and write to a whole class, and can only write to families whose children they take. Quick replies and attachments are still absent.
T21	Reports: attendance rate, marks distribution, homework rate, trend, at-risk	Implemented	
T22	Reports: "tell the family"	Implemented	403 (<code>notices/route.ts:121</code> vs <code>personas.ts:137-142</code>). Fixed at the persona: the family-notice path takes <code>notify-families</code> , which the teacher holds, rather than <code>reports create</code> , which would also have let a teacher write a school-wide notice.
T23	Parents' evening: view slots and bookings	Implemented	
T24	Parents' evening: open evening, release slot	Implemented	403 (<code>meetings/route.ts:93</code> vs <code>personas.ts:139</code>). Fixed at the persona: opening, booking and releasing take <code>book-meeting</code> , not <code>students edit</code> .
T25	Profile	Implemented	Read-only with field ownership shown.
T26	Settings: notifications, publishing rules, appearance, security, privacy	Partial	Two toggles work; rest <code>NotYetAvailable</code> . The stamped rows are gone; what is left is what the product stores.
T27	Help	Implemented	One false claim (audit trail), still there.
T28	Behaviour incidents, merits	Missing	Parked S-P.1.

#	Workflow	Status	Evidence and gap
T29	Seating plans, pupil notes, medical alerts in the register	Missing	Health data exists but the register roll does not show allergy or medical flags. Half of it is now in place: welfare is its own resource rather than a reading of boarding, and the teacher persona holds <code>schools.welfare.view</code> , so a class teacher is permitted to see an allergy. Nothing on the roll shows one yet.
T30	Leave requests, payslips, staff notices (non-teaching staff needs)	Missing	HR module has leave; no bridge for staff without an <code>Employee</code> link.
T31	HOD moderation queue in the portal	Missing	HOD is not pinned to any portal (<code>proxy.ts:59-65</code>); moderation lives in admin. <code>POR TAL_HOME_BY_ROLE</code> still has no HOD entry. The back-office rail now lands an HOD on moderation rather than on a page they cannot use, which makes the admin workspace bearable for them without giving them a portal home.

Counts: 15 implemented, 5 partial, none broken, 11 missing. At the audit date they were 11, 5, 2 and 13.

4. Benchmark gap

Graded against `../reference/k12-benchmark.md` §5, §6, §8, §10.

Capability	Benchmark norm	Huchu today	Priority
Register: take, submit, lock; lesson-by-lesson option; medical flags on the roll	Standard	Take and submit; locking is the office's by design; still no flags and still keyed by day	P1
Gradebook with weighting, custom columns, import from sheet	Standard	Fixed assessments from office; no columns	P1
Visible mark-sheet status and HOD feedback	Standard	Absent from the portal	P0
Report-card comments with comment bank	Standard	Absent	P1
Two-way messaging with parents, class broadcast	Standard	Reply, start and broadcast, all in-app; no quick replies or attachments	P2
Behaviour and merits	Standard	Parked	P2 (decision)
Cover request and staff absence	Common	Absent	P2
Resource upload	Standard	Link only	P1

Capability	Benchmark norm	Huchu today	Priority
Offline register	Standard in low-connectivity markets; sold on the site	Absent	P1
Shared-device safety (idle lock, sign out everywhere)	Common	Absent	P2
Own timetable, lesson plans, copy forward	Standard	Present	Done
Cohort analytics for the teacher	Uncommon; differentiator	Present	Done

5. Bugs and risks

#	Finding	Location	Severity
B1	Register never leaves DRAFT from the portal. No submit control; <code>me/attendance</code> only upserts DRAFT. Office board and parent app both read "not submitted". Fixed: saving returns the session id and whether the day can still be sent in, and the screen submits through the one route that owns the DRAFT to SUBMITTED transition. Saving and sending in stay two acts, because a register is taken in pieces.	<code>app/api/v2/schools/port al/teacher/me/attendance/route.ts:106-118; teacher-register-screen.tsx</code>	High
B2	Parents' evening open and release always 403 for TEACHER. Fixed: the route takes <code>book-meeting</code> .	<code>teacher-meetings-screen.tsx:205-260; meetings/route.ts:93; personas.ts:139</code>	High
B3	Reports "tell the family" always 403 for TEACHER. Fixed: the family-notice path takes <code>notify-families</code> .	<code>teacher-reports-screen.tsx:232-247; notices/route.ts:121; personas.ts:137-142</code>	High

#	Finding	Location	Severity
B4	Security: term-marks roll-up and read are permission-checked but not ownership-checked; a teacher can create a DRAFT sheet for a colleague's class and read any class's term marks. Same for <code>GET /assessments/[id]/scores</code> and <code>GET /assignments/[id]</code>. Fixed: one shared ownership check over the teaching assignment, asked by all three routes, with the roll-up needing every subject in the class because it rewrites the whole sheet. The office — SUPERADMIN, MANAGER, SCHOOL_ADMIN and REGISTRAR — reads every class deliberately: scoping a registrar by teaching assignment scopes them to nothing, and what stops the office writing marks is the persona grant, not this test.	<code>assessments/term-marks/route.ts:41-100</code> ; <code>assessments/[id]/scores/route.ts:48-100</code> ; <code>assignments/[id]/route.ts:55-63</code>	Medium
B5	Security and privacy: every teacher can read every office-addressed parent thread, and can start a thread with any guardian in the school, not only guardians of pupils taught. Fixed: the unassigned queue belongs to the office, a teacher sees the entries about children they take, the same rule governs answering as reading, and starting a thread is refused to a teacher who does not take the child.	<code>lib/schools/messages.ts:133-149, 201-210</code> ; <code>messages/route.ts:98-110</code>	Medium
B6	A teacher can lock their own register (<code>lock</code> is guarded by <code>attendance:submit</code>, which TEACHER holds), defeating lock as oversight. Fixed: locking needs a new <code>lock</code> verb, which the registrar and the office hold and no teacher does. The route also carried a second role test beside the grant, which this audit did not report: it admitted the bursar, who holds no attendance grant at all. It is gone.	<code>attendance/sessions/[id]/lock/route.ts</code>	Low

#	Finding	Location	Severity
B7	Stale copy: settings says parent messaging is not built while the Messages screen exists. Fixed: the sentence is gone and the screen it described is built.	teacher-settings-screen.tsx:283	Low
B8	Dead API surface: me/marks and handleTeacherPortalGet have no caller.	app/api/v2/schools/portal/teacher/me/marks/route.ts; app/api/v2/portal/_handlers.ts	Low
B9	HOD cannot reach the portal; PORTAL_HOME_BY_ROLE has no HOD entry and moderation additionally needs an isHod teacher profile with a class assignment.	proxy.ts:59-65; results/sheets/[id]/hod-approve/route.ts:44-58	Medium
B10	Service worker sits in front of /api/v2 on installed PWAs; e2e blocks it to get the portal past the skeleton.	public/sw.js; e2e/teacher-portal-shots.spec.ts:30-35	Medium
B11	Help promises an audit trail that no portal route writes. Still open. The one audit row added since — correcting a register that has already been sent in — is written by the office's session route, not by anything the portal calls, and reading a pupil record still writes nothing.	teacher-help-screen.tsx:222-228; lib/schools/audit.ts	Low
B12	Roadmap rows S-6.55, S-6.56, S-6.57, S-6.58 marked done with parts missing or broken. S-6.56 and S-6.57 are now true. S-6.55 (upload) and S-6.58 (five panels) are still overstated, and the rows themselves are unchanged.	roadmap	Doc
B13	Found while fixing B1, not by this audit: the rules deciding who may mark, submit and lock a register were written out inline in four routes, and the four disagreed. Fixed: three pure functions the routes ask, with the teacher screen reading the answer rather than comparing status strings.	lib/schools/register.ts:56-82; attendance/sessions/[id]/route.ts; attendance/sessions/[id]/lock/route.ts	Medium

#	Finding	Location	Severity
B14	Found during the implementation, not by this audit, and still open: a report card can only be printed while a publish window is open, and nothing outside the windows API ever writes one. At a school whose sheets are published and whose families can already read the marks in the portal, printing the report card throws. It is the office's screen rather than the teacher's, but it is the end of the chain the teacher's marks travel down.	lib/documents/schools-sources.ts:492, 549	Medium

6. Proposed edits

1. **Add "Submit register" (B1).** A second button after save that calls the existing `/attendance/sessions/[id]/submit`; keep save-as-draft for partial entry. Show the state chip (Draft, Submitted, Locked) on the register and on Today's "unmarked" list. Move `lock` to `attendance:edit` (B6). Done, with one change: locking went to a verb of its own rather than to `attendance:edit`, so that who may lock is stated directly rather than inherited from who may edit. The state chip is on the register; Today's list still does not carry it.
2. **Fix the two 403s (B2, B3) at the persona, not the button.** Add to TEACHER: `schools.students` action `book-meeting` (new verb, narrower than `edit`) and `schools.reports` action `notify-family` (new verb, narrower than `create`), then guard `POST /meetings` (open, release, book) and the family-notice path on those. Blanket `edit` and `create` would let a teacher edit pupils and write school-wide notices. Done; the verb is spelled `notify-families`, and it went to the bursar and the head of department as well, who were refused the same way on their own screens.
3. **Scope the roll-up and reads (B4).** Reuse the `ownedClassSubject` check from `me/lessons/route.ts:96-113` in `term-marks`, `scores` GET and `assignments/[id]` GET for non-moderators. Done, as one shared implementation, with the office exempt for the reason given in B4.
4. **Narrow message visibility (B5).** Office-queue threads visible only to the class teacher of the child named and to office roles; `start` restricted to guardians of pupils in the teacher's class subjects. Done.
5. **Surface sheet status in the marks book.** After roll-up show Draft / Submitted / Sent back with HOD note / Approved / Published, with Submit and Fix-and-resubmit calling the existing sheet routes and `me/marks`. This closes T9 without new backend work. Not done.
6. **Delete dead handlers (B8),** fix copy (B7, B11), correct roadmap rows (B12). B7 done; B8, B11 and the roadmap rows not.
7. **Scope the service worker (B10)** to the marketing and static routes, or register it only on the tenant workspace, until offline is designed. Not done.

7. Proposed restructuring

- **Rename the portal "Staff" in code and docs** to match the host and make room for HOD, warden and non-teaching staff homes. Route stays `/portal/teacher` until a migration is worth it, but the shell's role switch should not assume every user teaches.
- **HOD mode in the staff portal.** An HOD is a teacher with extra duties. Pin HOD to the staff portal, add a "Department" group to the rail with the moderation queue (reuse `moderation-queue-content.tsx` logic behind `me/department/sheets`) and the scheme-of-work editor the teacher screen already renders read-only. This removes the need for HODs to use the admin workspace.
- **Rail grouping.** The prototype groups Daily work (Today, Attendance, Enter marks, Marks book, Messages, Timetable), More (Lesson plans, Homework, Shared files, Reports, Meetings), Account. Keep it, but put an unread badge on Messages and a "to submit" count on Attendance.
- **Register keyed by lesson, not by day.** `SchoolAttendanceSession` is keyed by class, stream and date. Secondary schools take a register per period; the prototype's register screen is "Form 2A · Mathematics". Add an optional `timetableSlotId` to the session so subject registers are possible without breaking day registers.
- **Marks entry and Marks book are one screen** with a column picker, not two rail items, once S-6.44 lands.

8. Proposed new workflows and features

1. **Result-sheet lifecycle in the portal (T9):** status, HOD note, resubmit; notification to the teacher on HOD action.
2. **Custom assessment columns and weighting (S-6.44)** with the school's scheme as the default and a per-column override.
3. **Comment bank and report-card comments (T12):** `SchoolCommentBank` per subject and grade band; comments written into `SchoolResultLine.remarks` before submit; head's comment at publish.
4. **Messaging (S-7.1, S-6.47 to 49):** start thread with a family from the register or marks book, class broadcast, quick replies, attachment via `lib/uploads/`. Starting a thread and writing to a whole class are built, from the Messages screen rather than from the roll; quick replies, attachments and the entry points on the other screens are not.
5. **Resource upload (T16)** through the record files engine.
6. **Offline register (S-8.1):** register the staff portal in `lib/offline/module-registry.ts`; outbox the submit; resolve duplicates on `(classId, streamId, date)`.
7. **Medical and safeguarding flags on the roll (T29)** from `SchoolHealthRecord` (allergy, urgent gap) with restricted detail. The permission half of this is done: welfare is its own resource rather than a reading of boarding, so a day school has health records and a class teacher holds the view grant. The roll shows nothing yet.
8. **Cover request and staff absence (T18):** teacher reports absence with dates; office arranges cover using the existing `SchoolCoverAssignment`; the covering teacher sees the lesson plan.
9. **Behaviour and merits** if S-P.1 is un-parked: `SchoolBehaviourEvent` with category, points, sanction, parent visibility.

10. **Shared-device safety (S-6.59):** idle lock and sign out everywhere, using the session list the parent stories also need.

9. Open decisions

- Is the register per day or per lesson? The model says day; the prototype says lesson. Decide before offline work fixes the key. The register screen now shows the period and the time it belongs to, which is the prototype's reading of it, while the session it writes is still one a day.
- Who publishes marks to parents: the teacher under rules (S-6.46) or only the office (today)? The prototype has "Send marks to parents"; the roadmap says publishing is admin work.
- Does the report card keep its publish-window requirement (B14)? No school configures a window, so the requirement means the report card cannot be printed at all. The window deliberately does not gate the portal: that check was written into the parent marks route during the implementation and reverted, because it would have hidden every published mark from every family. The window gates the report card; the sheet's status gates the portal.
- Do HODs get a portal home, or stay in the admin workspace?
- Un-park behaviour and merits? Every teacher-facing benchmark product has it.